BOARD OF SECONDARY EDUCATION, KARACHI

SECONDARY SCHOOL (10TH CLASS) EXAMINATION, 2023

CHEMISTRY

Past Paper 2023

Retyped faithfully from the original board paper — same questions, no changes.

SECTION "A" MULTIPLE CHOICE QUESTIONS (MCQs)

Maximum Marks: 12

Note: NOTE: (i) Attempt all the Questions of this Section. (ii) Do not copy down the part questions. Write only the answer against the proper number of the Question and its part according to the question Paper. (iii) Each question carries 1 Marks

Q1. Choose the correct Answer for each from the given question:

Q1. The system will be stable in equilibrium when:

(A) Kc = Qc
(B) K < Qc
(C) K > Qc
(D) none of these

Q2. Corrosive effects on skin is caused by:

(A) acid
(B) base
(C) salt
(D) all of these

Q3. If pH value is greater than 7, then solution will be:

(A) Acidic
(B) Basic
(C) Netural
(D) Amphoteric

Q4. The functional group – COOH is used for:

(A) Alkyne
(B) Alcohol
(C) Phenol
(D) Carboxylic acid

Q5. The branch of Chemistry which deals with the hydrocarbon and their derivatives, is known as:

(A) Organic Chemistry
(B) Inorganic Chemistry
(C) Bio Chemistry
(D) Physical Chemistry

Q6. Glucose is:

(A) vitamin
(B) protein
(C) lipid
(D) carbohydrate

Q7. Amino acids are building block of:

(A) nucleic acid
(B) protein
(C) vitamin
(D) lipids

Q8. The centrifuge machine is used for the separation of:

(A) juice
(B) pH
(C) mud
(D) crystal

Q9. The altitude of stratosphere is:

(A) 40 to 45 km
(B) 50 to 55 km
(C) 60 to 65 km
(D) 70 to 75 km

Q10. Ozone is a gas that is found in the layer of:

(A) Troposphere
(B) Mesosphere
(C) Stratosphere
(D) Exosphere

Q11. Fresh water is present on the surface of earth:

(A) 0.3%
(B) 3%
(C) 0.2%
(D) 2%

Q12. The citric acid is used in the preparation of cold drinks for

(A) sweet taste
(B) bitter taste
(C) sour taste
(D) salty taste

SECTION "B" (Short Answer Question)

Maximum Marks: 24

> Note: NOTE: Answer any EIGHT questions from this section. Each question carries 3 marks.

Q2. Write a few points about society, technology & science.

Q3. The value of Kc for the reaction is 1×10^{-4}. $2HI(g) \rightleftharpoons H_2(g) + I_2(g)$. At a given temperature, the molar concentration of reaction mixture is, HI = 2×10^{-5} mol.dm^{-3}, $H_2 = 1 \times 10^{-5}$ mol.dm^{-3} and $I_2 = 1 \times 10^{-5}$ mol.dm^{-3}. Predict the direction of the reaction.

Q4. What are the limitations of Arrhenius's Theory.

Q5. Write three uses of salts.

Q6. Find pH, pOH, $[OH^-]$ and $[H^+]$ of 2.46×10^{-9} M KOH solution.

Q7. What do you mean by Vital Force Theory?

Q8. Write down the molecular, structural and condensed formulae of the following compounds. | Compound | Molecular formula | Structural formula | Condensed formula | | (i) Methane | | | | | (ii) Ethane | | | | | (iii) Propane | | | |

Q9. Define the following:

(i) Aldose

(ii) Ketose

(iii) Saccharides

Q10. Elaborate Amino acids as monomers or building black of protein.

Q11. How can we remove temporary hardness of water by Clark's method.

Q12. Write three characteristics of equilibrium constant Kc.

Q13. Give reason of the following:

(i) Petroleum is "Black Gold"

(ii) Alkanes said to be saturated Hydrocarbons.

Section C (Detailed –Answer Question)

Maximum Marks: 24

> Note: NOTE: Attempt any FOUR questions from this Section. Each question tries (Six) 6 marks.

Q14. Explain the process of sugar preparation from sugarcane

Q15. What is high performance liquid chromatography (HPLC)? Explain it.

Q16. Distinguish between fat soluble and water soluble vitamin.

Q17. Differentiate between saturated and unsaturated hydrocarbons.

Q18. Define buffers. What is its composition? How are these utilized?

Q19. Describe Global warming.